Romain Lhoussaine

R&D Engineer - Scientific Software & Space Image Processing

0661783051 romain.lhoussaine@gmail.com www.linkedin.com/in/lhoussaineromain Toulouse, 31000

PROFESSIONAL EXPERIENCE

R&D Engineer: Hyperspectral Image Processing (HYP4U)

ORUS

2026 - Present Toulouse, France

Developing L0–L2 processing algorithms for the HYP4U hyperspectral mission:

- Spectral correction pipeline (Fabry-Pérot effect), synthetic spectrum simulation, and ATBD documentation
- Computation optimisation and parallelisation

- *Hyperspectral, L0-L2, Python, Sensors, Linescan, numpy, scipy, lmfit*

Satellite Engineer: FlatSat & Ground Segment (ENDURANCE)

Infinite Orbits

2024 - 2025 Toulouse, France

Development and integration of test benches for the in-orbit servicing program:

- Receipt and integration of flight hardware and software elements
- Development of software tools for FlatSat operations
- Unit testing and validation of flight software
- Integration testing: TC/TM command dispatch and subsystem validation

- *ECSS, Test Bench, Flight Software, Python, Yamcs, Unix Shell, DP900 Series Programming*

R&D Engineer: Scientific Data Pipelines (Sentinel)

ACRI-ST

2024 Toulouse, France

Developed scientific data processing chains for remote sensing earth's observation:

- Re-engineering of data processors for the Sentinel-3 SYNERGY instrument
- Implementation of the L2 pipeline processor for Sentinel-8 Land Surface Temperature Monitoring (LSTM)

- *Dask, GitLab Pro, Agile, CI/CD, Docker, Kubernetes, Devpod, Rancher, xarray*

R&D Engineer: Space Instrumentation (EUCLID/NISP)

Marseille Particle Physics Center

2023 - 2024 Marseille, France

In-flight commissioning of the NISP instrument aboard the EUCLID space telescope:

- Verification and monitoring of in-flight performance (spectroscopy, photometry, calibration)
- Development of data analysis code and validation of calibration procedures
- Technical support to the Instrument Scientist

- *Jupyter Notebooks, Spectroscopy, Satellite, AI, Astrophysics, Dask, Cluster, Fits*

R&D Engineer: Instrumentation & Data Analysis (PFS)

Marseille Astrophysics Laboratory

2021 - 2023 Marseille, France

https://doi.org/10.1117/12.2628152

Simulation and data analysis for spectrographic alignment of optics and spectral channels of the Prime Focus Spectrograph (PFS) dedicated to the Subaru ground-based telescope.

- *Cluster, Instrumentation, Data Reduction Pipeline, Image Processing*

R&D Engineer: Radiographic Image Processing

ArianeGroup

2020 Le Haillan, France

Developed a 3D image processing algorithm for automated detection of ripples in tomography-imaged composite parts:

- Designed a radiographic processing chain for non-destructive testing automation of carbon composite pieces

- *Image Processing Pipeline, Eclipse, Hybrid Programming C/Java, JNI/JNA, CPU Parallelization, Big Data, ImageJ, Optimization*

EDUCATION

Master 2 OSAE: Astronomy and Space Systems Tools

Paris Observatory / PSL University

09/2019 - 10/2020 Meudon, (92)

- *Advanced Numerical Methods, Machine Learning, Embedded Systems, Computer Projects, Signals and Systems, Optics, Electronics, Finite Elements*

Master 1 SUTS: Universe Sciences and Space Technologies

Paris Observatory / PSL University

09/2018 - 07/2019 Meudon, (92)

- *Galaxy Physics, The Milky Way and Local Group, Dynamics of Rotating Bodies, Cosmology, Associated Instrumentation and Methods, Signal Processing and Inverse Methods, Scientific Programming, Optics and Imaging*

Dual Bachelor's Degree in Physics & Chemistry

University Paris-Sud / Paris-Saclay

09/2016 - 07/2018 Orsay, France (91)

- *Astrophysics, Waves and Vibrations, Mathematics for Sciences, Linear and Geometric Optics, Fluid and Solid Mechanics*

SKILLS

Proficient

Python | C | CI/CD | Dask | Bash | Unix Shell | Git

Intermediate

C++ | Fortran 95 | SQL | IDL | R | Docker | Kubernetes

Beginner

Rust | C# | Java

LANGUAGES

English	Fluent
Spanish	Professional
French	Native

HOBBIES

Volunteer Firefighter

2013-2023